

DevOps Engineer

Step 5: Master Cloud Platforms

CareerCompass • Detailed Concept Notes • Research-backed learning guide

Concept & Learning Objective

Cloud engineering is not simply renting a server. Learn the service categories and the design trade-offs behind them: compute, storage, networking, IAM, managed databases, monitoring, availability and cost.

Pick one major cloud first. The concepts transfer across providers, but service names and operational details differ. Use least privilege and cost controls from the beginning. Infrastructure as Code should eventually make the environment reproducible.

Core Concepts — Detailed Breakdown

- Compute: VMs, containers, serverless and managed application platforms.
- Storage: object, block and file storage; match storage type to access pattern.
- Networking: virtual networks, subnets, routing, security groups/firewalls and load balancing.
- IAM: identities, roles/policies and least privilege.
- Regions/AZs: geographic and fault-domain considerations.
- Managed services: reduce operational burden but introduce provider-specific dependencies.
- Observability: logs, metrics, traces and alerts.
- Cost: usage-based billing, budgets, idle resources and right-sizing.
- IaC: define infrastructure declaratively and review changes like application code.

What You Should Be Able To Do

- Explain IaaS/PaaS/SaaS.
- Deploy one small application.
- Create least-privilege access.
- Configure a budget alert.
- Read logs and basic metrics.
- Draw the application's cloud architecture.

Hands-on Project / Practice

- Deploy a containerized application to one cloud. Document compute, network, IAM, storage, monitoring and estimated monthly cost. Destroy the environment safely after practice if it is not needed.

Recommended References

- AWS Skill Builder — <https://skillbuilder.aws/>
- Terraform Documentation — <https://developer.hashicorp.com/terraform/docs>
- AWS Well-Architected Framework — <https://docs.aws.amazon.com/wellarchitected/latest/framework/>

Note: These notes are designed to teach the concepts and provide a practical learning path. Always prefer the linked official documentation for current APIs, SDK versions and platform-specific release requirements.